

Mani Majd

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EDUCATION

University of Toronto

Bachelor of Applied Science (B.A.Sc) in Engineering Science + PEY Co-op

Toronto, ON

Sep 2024 - April 2029

- Robotics major + AI minor
- Relevant Courses: Computer Programming (Python), Computer Algorithms and Data Structures (C), Digital and Computer Systems (RISC-V), Calculus I, II & III, Probability and Statistics, Linear Algebra
- Cumulative GPA: 3.91, Dean's Honours List (×4)

EXPERIENCE

Undergraduate Researcher

LCRAIN Lab, York University

Toronto, ON

April 2025 - August 2025

- Designed and implemented spiking neural networks (SNNs) for energy-efficient visual perception in autonomous driving applications using PyTorch and snnTorch, leveraging surrogate-gradient learning for spatiotemporal backpropagation.
- Developed data preprocessing, augmentation, annotation, and spike-encoding pipelines to convert road-surface imagery into event-driven representations for neuromorphic learning.
- Designed, trained, and optimized an SNN-based pothole detection and classification model, achieving 97.7% F1 score, 98.0% recall, and 97.7% accuracy while demonstrating robust generalization across diverse road conditions.
- Evaluated model performance using confusion matrices, ROC analysis, and spike-activation visualization, while investigating deployment on Intel Loihi and FPGA-based accelerators for real-time, low-power edge AI inference.

Robotics Research Intern

Technical University of Munich, Learning Systems and Robotics Lab

Toronto, ON

May 2026 - August 2026

- Collected large-scale human and robotic demonstration datasets through teleoperation to train and evaluate imitation learning policies for robotic manipulation tasks.
- Trained diffusion-based and vision-language-action policies, investigating a hybrid learning framework that leverages abundant human demonstrations alongside limited robot demonstrations to reduce the cost of robot data collection while maintaining competitive task performance.
- Designed a temporal reasoning pipeline for semantic object search in semi-static indoor environments by modeling historical object-placement patterns and habitual object locations.
- Developed a Bayesian probabilistic search framework that combines temporal priors, environment-specific anchors, and LLM-based commonsense reasoning to infer object-location probabilities from natural language queries and generate efficient search policies.

PROJECTS

Predicting the Remaining Useful Life of Jet Turbofan Engines Using Neural ODEs

December 2025

- Developed Neural Ordinary Differential Equation (NODE) and CNN–NODE architectures in PyTorch for remaining useful life prediction from multivariate time-series sensor data using NASA's CMAPSS benchmark dataset.
- Proposed a continuous-depth latent-state modeling framework integrating CNN feature extraction with Neural ODE dynamics, outperforming multiple state-of-the-art baselines on fixed operating-condition benchmarks.
- Published at the 2026 IEEE International Conference on Prognostics and Health Management (ICPHM).

Deep Neural Network for Image Classification

July 2025

- Developed an L-layer deep neural network from scratch in Python to classify RGB images as cat or non-cat, leveraging NumPy for matrix operations and numerical computations.
- Designed the model architecture as a stack of linear layers and ReLU activations, followed by a sigmoid output layer to predict binary class probabilities.
- Manually implemented all core neural network components, including forward and backward propagation, activation functions (ReLU, Sigmoid), and binary cross-entropy loss optimization using gradient descent.
- Trained and evaluated the model on a labeled image dataset, improving classification accuracy over a logistic regression baseline through effective parameter tuning and data preprocessing.

Weather Forecasting

November 2024

- Developed a LSTM-based time-series model in PyTorch to forecast Toronto weather variables using multi-step prediction.
- Built a sliding-window data generator that converted raw time-series into supervised samples with a 10-step lookback and 6-step forecast horizon, enabling multi-step forecasting across multiple targets.
- Implemented a two-layer LSTM architecture with dropout regularization and a reshaped linear output head to improve forecast stability.

EXTRACURRICULARS

University of Toronto Machine Intelligence Team (UTMIST)

Toronto, ON

Machine Learning Project Lead

Sep 2025 – June 2026

- Led a 5-member team developing OCT-EyeSNN, an end-to-end spiking neural network framework for classifying retinal OCT scans into four clinically relevant disease categories.
- Designed and implemented the full ML pipeline, including dataset preprocessing, spike encoding, unsupervised pretraining, and surrogate-gradient optimization for a convolutional SNN, achieving 97.7% accuracy on the held-out test set.
- Built a benchmarking and neuromorphic evaluation pipeline comparing SNNs against compact CNN and Vision Transformer baselines, while evaluating deployment on Intel Loihi simulation for energy efficiency, latency, and memory footprint.

University of Toronto Formula Racing (UTFR)

Toronto, ON

Deep Learning Perception Member

May 2025 – October 2025

- Contributed to the Deep Learning Perception Team developing Canada's first driverless Formula SAE (FSAE) car, focused on enabling autonomous cone detection for track navigation.
- Trained and fine-tuned YOLO object detection models (v8, v10, v11) on the team's cone detection dataset, optimizing training pipelines through data preprocessing, augmentation, and hyperparameter tuning.
- Evaluated detection performance using key metrics (IoU, mAP50–95, precision, recall) and improved post-processing algorithms like Non-Maximum Suppression (NMS) to refine predictions.

University of Toronto Robotics Association (UTRA)

Toronto, ON

Software Team Member – Perception subdivision

Sep 2024 – Sep 2025

- Calibrated a ZED stereo camera with OpenCV for depth-based terrain mapping, detecting ramps and uneven surfaces.
- Developed a lane detection pipeline using a YOLO object detection model, enabling the rover to identify lane boundaries for autonomous path tracking.
- Optimized a YOLO-based segmentation model through dynamic quantization in PyTorch, reducing model size and improving real-time inference speed on embedded systems.
- Tuned training pipelines, including data augmentation, annotation refinement, and hyperparameter adjustments, to improve object detection accuracy in challenging outdoor environments.

SKILLS

Programming/Software: Python, C, MATLAB, Verilog, Git

Machine Learning/AI: PyTorch, snnTorch, OpenCV, NumPy, pandas, Matplotlib, model quantization & optimization, YOLO

Embedded Systems: ROS, Zephyr RTOS, stereo vision (ZED)

Data Analysis/Modelling: Numerical Methods, Signal Processing, Statistical Analysis